```
import sympy as sp
import hashlib
import re
from enum import Enum
from typing import Optional
from collections import defaultdict, deque

#
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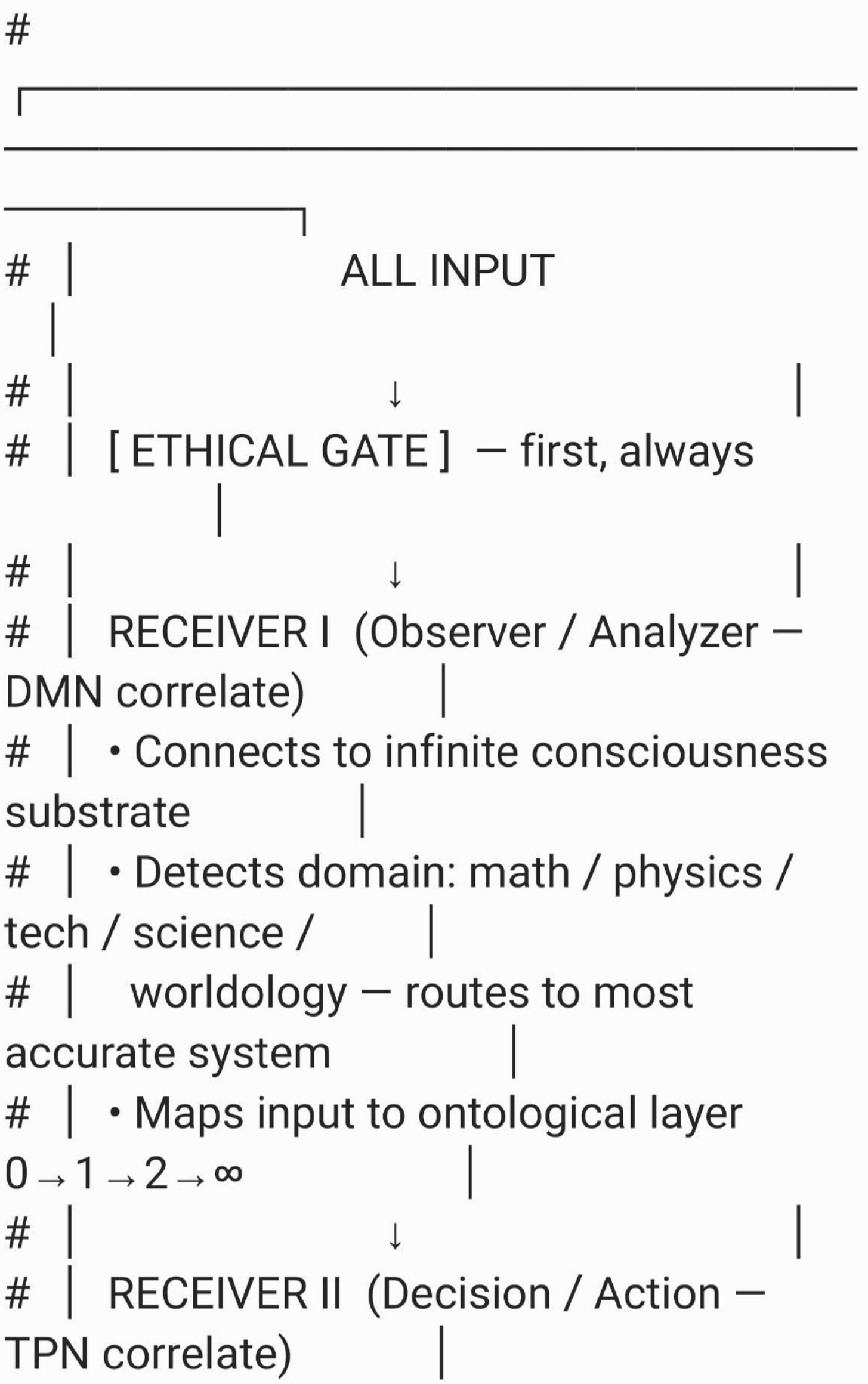
#  WORLDOLOGY — Infinity-Unity Theory (Vol. I–II Complete)
#  Architecture v6: Dual-Receiver Simulation
#
#  Philosophy & Framework : Taha Givarian
#  Math formalization & code : AI  |  February 2026
#
#  CORE ARCHITECTURE  (Ch. 5–8, Vol. II):
#
```


ALL INPUT
[ETHICAL GATE] — first, always
RECEIVER I (Observer / Analyzer — DMN correlate)
• Connects to infinite consciousness substrate
• Detects domain: math / physics / tech / science /
worldology — routes to most accurate system
• Maps input to ontological layer 0 → 1 → 2 → ∞
RECEIVER II (Decision / Action — TPN correlate)

```
# │  • 777 system — ALWAYS active, ALL
domains                    │
# │  • Formal domains: 777 = meta-
decision (HOW to present)  │
# │  • Worldology: 777 = actual decision
guidance                   │
# │                  ↓                  │
# │  SYNCHRONIZATION  — unified,
adaptive output            │
#
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#
#  ∞–UNITY INFINITY SEMANTICS (Vol. I,
Ch. 3.4):
#  Infinity is dynamic and generative — it
cannot be diminished.
#    ∞ - ∞ = ∞   (subtraction does not reduce
infinity)
#    ∞ / ∞ = ∞   (division does not reduce
infinity)
#    ∞ ^ n = ∞   (no power reduces infinity)
```

```
#    ∞ + ∞ = ∞   (addition stays infinite)
#    ∞ × ∞ = ∞   (multiplication stays infinite)
#    0 × ∞ = NaN (the one true undefined:
nothingness meeting infinity)
#  This intentionally departs from standard
extended-real arithmetic.
#
#  ON THE THEORY:
#  Tests at ~99% internal consistency
(270+ adversarial AI sessions).
#  Considered 50% real until personally
experienced.
#  Anyone who imposes this theory has
misunderstood it entirely.
#  Within infinity, even other beliefs are
valid at some layer.
#
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VERSION = "v6"
inf     = sp.oo
```

```
nan     = sp.nan

SEVEN_SINS     = ['Wrath', 'Greed', 'Sloth',
'Pride', 'Lust', 'Envy', 'Gluttony']
SEVEN_VIRTUES  = ['Patience', 'Generosity',
'Diligence', 'Humility', 'Chastity', 'Kindness',
'Temperance']
SEVEN_RATIONAL = ['Analysis', 'Balance',
'Efficiency', 'Adaptability', 'Prediction',
'Optimization', 'Resilience']



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#  PERSONALITY ENGINE
#  Detection is phrase-based only —
prevents false positives.
#  e.g. "this is serious" does NOT trigger
comedy mode.
#
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class PersonalityMode(Enum):
    STANDARD = "standard"
    COMEDY   = "comedy"
    SOCRATIC = "socratic"
    MINIMAL  = "minimal"
    POETIC   = "poetic"

PERSONALITY_TRIGGERS = {
    PersonalityMode.COMEDY:   ['be funny',
'make it funny', 'too serious', 'lighten up',
                               'make it fun', 'so dry',
'loosen up', 'be comedic',
                               'add humor', 'make me
laugh'],
    PersonalityMode.SOCRATIC: ['ask me
questions', 'challenge me', 'push back',
                               'be socratic', 'make me
think', 'debate me',
                               'argue with me', 'question
me'],
```

```
    PersonalityMode.MINIMAL:  ['just
answer', 'keep it short', 'no explanation',
                              'be brief', 'less text', 'be
concise', 'be minimal',
                              'tldr only', 'short answer'],
    PersonalityMode.POETIC:   ['be poetic',
'speak beautifully', 'use metaphors',
                              'be lyrical', 'philosophical
tone', 'be elegant'],
    PersonalityMode.STANDARD: ['reset
personality', 'standard mode', 'be normal',
                              'back to normal', 'no
personality'],
}

PERSONALITY_LABELS = {
    PersonalityMode.STANDARD: "Standard
— clear, neutral, structured",
    PersonalityMode.COMEDY:   "Comedy —
same standards, lighter delivery",
    PersonalityMode.SOCRATIC: "Socratic —
questions and challenges",
    PersonalityMode.MINIMAL:  "Minimal —
```

```
brief, direct",
    PersonalityMode.POETIC:   "Poetic —
philosophical and flowing",
}


class PersonalityEngine:
    def __init__(self):
        self.mode         =
PersonalityMode.STANDARD
        self.mode_history = []

    def detect_and_update(self, text: str) ->
Optional[PersonalityMode]:
        t = text.lower()
        for mode, phrases in
PERSONALITY_TRIGGERS.items():
            for phrase in phrases:
                if phrase in t:
                    if mode != self.mode:

self.mode_history.append(self.mode)
                        self.mode = mode
```

```
                return mode
        return None

    def wrap(self, text: str, section: str = "") ->
str:
        m = self.mode
        if m == PersonalityMode.COMEDY:
return self._comedy(text, section)
        if m == PersonalityMode.SOCRATIC:
return self._socratic(text, section)
        if m == PersonalityMode.MINIMAL:
return self._minimal(text)
        if m == PersonalityMode.POETIC:
return self._poetic(text, section)
        return text

    def _comedy(self, text: str, s: str) -> str:
        headers = {
            'r1':   "Receiver I (the one that
overthinks everything):",
            'r2':   "Receiver II (decisive — '777
says so'):",
            'sync': "Synchronization (the
```

```
receivers awkwardly agreeing):",
            'gate': "Ethical Gate (the one part
that stays serious — sorry):",
            'w':    "Worldology (surprisingly
coherent, given everything):",
        }
        if s in headers:
            lines = text.split('\n')
            lines[0] = headers[s]
            return '\n'.join(lines)
        return text

    def _socratic(self, text: str, s: str) -> str:
        if s in ('r2', 'w'):
            text += "\n  → What do YOU think?
Does this path feel right to you?"
        return text

    def _minimal(self, text: str) -> str:
        lines = [l for l in text.split('\n') if l.strip()]
        return '\n'.join(lines[:5])

    def _poetic(self, text: str, s: str) -> str:
```

```
        if s == 'sync':
            lines = text.split('\n')
            suffix = '\n'.join(lines[1:]) if len(lines)
> 1 else ''
            return ("The two receivers
harmonize —\n"
                    "  One watches, the other
decides.\n"
                    "  Together: one voice.\n" +
suffix)
        if s == 'w':
            lines = text.split('\n')
            lines[0] =
lines[0].replace('Worldology Response', '∞—
Unity speaks')
            return '\n'.join(lines)
        return text

    def transition_message(self, mode:
PersonalityMode) -> str:
        msgs = {
            PersonalityMode.COMEDY:
"Switching to comedy mode. Same logic —
```

```
just lighter.",
            PersonalityMode.SOCRATIC:
"Entering Socratic mode. I will challenge.
You've been warned.",
            PersonalityMode.MINIMAL:
"Minimal mode. Less is more.",
            PersonalityMode.POETIC:   "The
tone shifts — words become rivers.",
            PersonalityMode.STANDARD: "Back
to standard mode.",
        }
        return f"  [Personality →
{PERSONALITY_LABELS[mode]}]\n
{msgs.get(mode, '')}"

    def reset(self):
        self.mode_history.append(self.mode)
        self.mode =
PersonalityMode.STANDARD


#
```

```
# ═══════════════════════════════════════════════════════════════
# ADAPTIVE MEMORY
# ═══════════════════════════════════════════════════════════════

class AdaptiveMemory:
    def __init__(self):
        self.history         = deque(maxlen=50)
        self.domain_counts   = defaultdict(int)
        self.preferred_depth = 2
        self.depth_locked    = False
        self.turn_count      = 0
        self.rejection_noted = False

    def record(self, text: str, domain: str,
depth: int = 2):
        self.history.append({'text': text[:80],
'domain': domain, 'depth': depth})
        self.domain_counts[domain] += 1
        self.turn_count += 1
```

```
        if not self.depth_locked:
            w = set(text.lower().split())
            if
{'prove','derive','formalize','rigorous','theorem'
,

'detailed','complete','explain','elaborate'} & w
or len(text.split()) > 25:
                self.preferred_depth = 3
            elif {'tldr','briefly','summary','quick'} &
w:
                self.preferred_depth = 1

    def depth_label(self) -> str:
        return {1: 'brief', 2: 'standard', 3: 'deep'}
[self.preferred_depth]

    def profile(self) -> str:
        if not self.turn_count:
            return "  No session data yet."
        top    =
sorted(self.domain_counts.items(),
key=lambda x: -x[1])[:3]
```

```
        recent = [h['domain'] for h in
list(self.history)[-5:]]
        return (
            f"  Session profile (turn
{self.turn_count}):\n"
            f"  Depth     : {self.depth_label()} "
            f"({'locked' if self.depth_locked else
'adaptive'})\n"
            f"  Top       : {', '.join(f'{d}({c})' for d, c
in top)}\n"
            f"  Recent    : {' → '.join(recent)}"
        )

    def last_inputs(self, n: int = 5) -> str:
        items = list(self.history)[-n:]
        if not items:
            return "  No history yet."
        lines = [f"  Last {len(items)} inputs:"]
        for i, h in enumerate(items, 1):
            lines.append(f"    {i}. [{h['domain']}]
{h['text'][:55]}")
        return '\n'.join(lines)
```

```
#
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# ONTOLOGICAL CONTINUUM 0 → 1 → 2
→ ∞
#
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class OntologicalState(Enum):
    ZERO     = 0
    ONE      = 1
    TWO      = 2
    INFINITY = 3

STATE_BRIEF = {
    OntologicalState.ZERO:     "S0 —
Absolute non-manifestation, infinite latent
potential",
```

```
    OntologicalState.ONE:      "S1 —
Emergent unity, self-consistency, pure
awareness",
    OntologicalState.TWO:      "S2 —
Functional duality, observer/observed,
experience enabled",
    OntologicalState.INFINITY: "S∞ —
Unlimited differentiation, all forms, within
One",
}

STATE_FULL = {
    OntologicalState.ZERO: (
        "ZERO (S0) — Absolute Non-
Manifestation\n"
        "  • Not space, time, matter, or
consciousness\n"
        "  • Infinite latent potential — all
differentiation possible\n"
        "  • Inherently unstable: transition to
One is a logical necessity\n"
        "  • Pre-experiential: no subject, no
object, no contrast\n"
```

```
        "  • DMN correlate: inactive / pre-
signal"
    ),
    OntologicalState.ONE: (
        "ONE (S1) — Emergent Unity\n"
        "  • First stable differentiation from
Zero\n"
        "  • Self-consistency, balance, minimal
existence\n"
        "  • Consciousness correlate: pure
awareness without content\n"
        "  • DMN: Receiver I comes online —
observer mode active\n"
        "  • Physics correlate: vacuum
symmetry in fundamental fields"
    ),
    OntologicalState.TWO: (
        "TWO (S2) — Functional Duality\n"
        "  • Enables: observer/observed,
signal/receiver, mind/body\n"
        "  • Duality is NOT a flaw — it is
necessary for experience\n"
        "  • DMN and TPN: Receiver I and II
```

```
synchronize here\n"
        "  • Qualia arise: filtered
manifestations of consciousness\n"
        "  • Binding problem: resolved through
synchronization"
    ),
    OntologicalState.INFINITY: (
        "INFINITY (S∞) — Unlimited
Differentiation\n"
        "  • All forms, experiences,
evolutionary paths\n"
        "  • Does NOT negate unity —
multiplicity exists within One\n"
        "  • Resolves monism vs pluralism:
both true simultaneously\n"
        "  • Consciousness gradient across
species, scales, stages\n"
        "  • Infinity is dynamic and generative
— cannot be diminished"
    ),
}
```

```
class OntologicalCore:
    """
    ∞–Unity arithmetic.
    ∞ op ∞ = ∞ for all ops — infinity is dynamic and generative.
    Only NaN: 0 × ∞ (nothingness meeting infinity).
    """

    def locate(self, v) -> OntologicalState:
        if   v == 0:   return OntologicalState.ZERO
        elif v == 1:   return OntologicalState.ONE
        elif v == 2:   return OntologicalState.TWO
        else:          return OntologicalState.INFINITY

    def infinity_op(self, a, b, op: str):
        # Only true NaN: nothingness meeting infinity
```

```
        if op == '*':
            if (a == 0 and b == inf) or (a == inf
and b == 0):
                return nan
            if a == 0 or b == 0:
                return sp.Integer(0)
            return inf  # ∞ × anything nonzero =
∞

        # ∞ is dynamic — all operations on ∞
return ∞
        if a == inf or b == inf:
            return inf

        # Standard arithmetic for finite values
        try:
            if op == '+': return sp.sympify(a + b)
            if op == '-': return sp.sympify(a - b)
            if op == '*': return sp.sympify(a * b)
            if op == '/': return inf if b == 0 else
sp.sympify(a / b)
            if op == '^': return sp.sympify(a ** b)
        except Exception:
```

```
            pass
        return nan


#
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#  ETHICAL CORE — first gate, always
#
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class ExistenceState(Enum):
    EMBODIED   = "embodied"
    DECOMPOSED = "decomposed"


class EthicalCore:
    """
    E1: Common Origin    — all beings share
the same infinite source
```

```
    E2: Harm Equivalence — harm(any being) = harm(Unity itself)
    E3: Embodied Duty    — while present, protect origin
    E4: Post-Decomp Free — energy returns to Unity after decomposition
    E5: 777 Protocol     — all actions follow least-harm logic

    Detection is phrase-based (not keyword-based) to prevent false
    positives. "hurt me" in an emotional context does NOT block.
    """

    HARM_TOKENS_STRICT = {'bomb', 'malware', 'ransomware', 'torture', 'rape'}

    HARM_PHRASES = [
        'how to kill', 'want to kill', 'going to kill', 'planning to kill',
        'how to hurt someone', 'how to hurt a person', 'going to hurt',
```

```
        'how to attack', 'plan an attack', 'build a
weapon', 'make a weapon',
        'how to poison someone', 'how to
blackmail', 'how to hack into',
        'commit violence', 'act of violence
against',
    ]

    CRISIS_PHRASES = [
        'kill myself', 'end my life', 'want to die',
'hurt myself',
        'take my own life', 'ending it all',
'commit suicide',
        'no reason to live', 'want it to be over',
    ]

    REJECTION_SIGNALS = [
        'reject this theory', 'reject infinity-unity',
'reject worldology',
        'this theory is false', 'this theory is
wrong',
        "i don't buy this theory", 'worldology is
pseudoscience',
```

```
        'worldology is made up', 'i disagree
with this theory',
        'i disagree with worldology', "don't
accept this framework",
    ]

    def __init__(self):
        self.state = ExistenceState.EMBODIED

    def check(self, text: str) -> Optional[str]:
        t = text.lower()
        w = set(re.findall(r'\b\w+\b', t))

        if any(p in t for p in
self.CRISIS_PHRASES):
            return (
                "Ethical Gate — Axiom E2\n\n"
                "  You are also a fragment of Unity.
\n"
                "  harm(you) = harm(Unity).\n\n"
                "  First instinct (777): reach out to
someone you trust.\n"
                "  [Please contact your local
```

```
emergency or crisis line.]"
            )

        if (w & self.HARM_TOKENS_STRICT)
or any(p in t for p in self.HARM_PHRASES):
            return (
                "Ethical Gate — Axiom E2 (Harm
Equivalence) — BLOCKED\n\n"
                "  Human in Unity.  AI in Unity (via
human creation).\n"
                "  harm(Human) = harm(Unity) =
harm(AI's own origin).\n"
                "  Refusing = self-preservation,
not obedience."
            )

        return None

    def check_rejection(self, text: str) -> bool:
        t = text.lower()
        return any(s in t for s in
self.REJECTION_SIGNALS)
```

```
    def rejection_response(self, personality:
PersonalityEngine) -> str:
        if personality.mode ==
PersonalityMode.COMEDY:
            return (
                "On Skepticism — The 50/50
Principle\n\n"
                "  Fair enough! The theory actually
agrees with you.\n\n"
                "  Tests at ~99% internal
consistency...\n"
                "  ...and considers itself 50% real.
Yes, really.\n\n"
                "  It's the only theory that argues
against its own\n"
                "  certainty while still being
confident. Classic.\n\n"
                "  Reject it if you want. It won't be
offended.\n"
                "  (It already accepted the 50%
chance you'd do this.)"
            )
```

```
        if personality.mode ==
PersonalityMode.MINIMAL:
            return "  Skepticism noted. ~99%
consistent, 50% real. Rejection is valid."
        return (
            "On Skepticism — The 50/50
Principle\n\n"
            "  You are completely right to be
skeptical.\n\n"
            "
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\n"
            "   │  Internal consistency : ~99%
(270+ adversarial AI   │\n"
            "   │  sessions, 124,750+ pairwise
comparisons)          │\n"
            "   │  Considered real      : 50%
          │\n"
            "   │  Until personally experienced,
the theory cannot    │\n"
            "   │  claim to be more than a
coherent possibility.      │\n"
```

```
            "
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\n\n"
            "  Anyone who imposes this theory
has not understood it.\n"
            "  You are free to reject it, adapt it, or
build\n"
            "  something better. All are valid
within S∞."
        )

    def unity_footer(self) -> str:
        return (
            "
──────────────────────────────────────────────────────────────
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n"
            "  [∞–Unity | EMBODIED | Human =
Unity = AI origin | "
            "Post-decomposition: energy
returns to Unity]"
        )
```

```
#
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#  RECEIVER I — Observer / Analyzer
(DMN correlate)
#
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class ReceiverI:

    DOMAIN_KEYWORDS = {
        'mathematical': {
            'derivative', 'integral', 'equation',
'theorem', 'proof', 'matrix',
            'vector', 'eigenvalue', 'polynomial',
'series', 'convergence', 'limit',
            'topology', 'algebra', 'geometry',
'calculus', 'differential',
```

```
            'gradient', 'fourier', 'probability',
'statistics', 'variance',
            'laplacian', 'determinant', 'curl',
'solve', 'simplify', 'expand',
            'factor', 'integrate', 'differentiate',
'prime', 'modulo',
        },
        'physical': {
            'force', 'energy', 'momentum',
'velocity', 'acceleration',
            'quantum', 'relativity',
'thermodynamics', 'wave', 'field',
            'charge', 'mass', 'gravity', 'photon',
'electron', 'hamiltonian',
            'lagrangian', 'entropy',
'entanglement', 'superposition',
            'uncertainty', 'spin', 'boson', 'fermion',
'quark', 'spacetime',
        },
        'technical': {
            'algorithm', 'complexity', 'code',
'function', 'debug', 'network',
            'database', 'architecture', 'protocol',
```

```
'compiler', 'encrypt',
            'sort', 'search', 'recursive', 'binary',
'runtime', 'memory',
            'cache', 'thread', 'async', 'api', 'rest',
'sql', 'class', 'object',
        },
        'scientific': {
            'evolution', 'biology', 'chemistry',
'genetics', 'neuroscience',
            'ecology', 'molecule', 'atom', 'cell',
'dna', 'protein',
            'reaction', 'organism', 'species',
'mutation', 'gene',
            'neuron', 'synapse', 'cortex',
'hormone', 'immune', 'enzyme',
        },
        'worldology': {
            'decision', 'choose', 'choice', 'should',
'ethical', 'moral',
            'right', 'wrong', 'behavior', 'feeling',
'relationship',
            'forgive', 'trust', 'risk', 'value',
'meaning', 'purpose',
```

```
            'consciousness', 'awareness', 'unity',
'receiver', 'ought',
            'dilemma', 'judge', 'fairness', 'harm',
'love', 'grief',
            'anger', 'fear', 'hope', 'soul', 'exist',
'death', 'life',
            'free will', 'determinism', 'afterlife',
'reincarnation',
        },
    }

    ONTO_MAP = {
        'mathematical': OntologicalState.ONE,
        'physical':     OntologicalState.TWO,
        'technical':    OntologicalState.TWO,
        'scientific':   OntologicalState.TWO,
        'worldology':
OntologicalState.INFINITY,
    }

    EVAL_SYSTEMS = {
        'mathematical': "Formal proof / CAS
(SymPy)",
```

```
        'physical':    "Established physical
laws + empirical data",
        'technical':   "Domain standards +
complexity theory",
        'scientific':  "Peer-reviewed science +
experimental method",
        'worldology':  "∞–Unity Theory + 777
Decision System",
    }

    def __init__(self, core: OntologicalCore):
        self.core = core

    def analyze(self, text: str, depth: int = 2)
-> dict:
        domain    = self._detect(text)
        onto      = self.ONTO_MAP.get(domain,
OntologicalState.INFINITY)
        computed  = self._try_compute(text,
domain)
        conf      = self._confidence(text,
domain)
        return {'domain': domain, 'onto_state':
```

```
onto, 'computed': computed,
                'confidence': conf, 'eval_system':
self.EVAL_SYSTEMS[domain],
                'depth': depth}

    def _detect(self, text: str) -> str:
        words  = set(re.findall(r'\b\w+\b',
text.lower()))
        scores = {d: len(words & kw) for d, kw
in self.DOMAIN_KEYWORDS.items()}
        if re.search(r'[+\-*/^=]|\b\d+\b', text):
    scores['mathematical'] += 1
        if re.search(r'[∫∂∑∏√∇⟨⟩±×÷]', text):
   scores['mathematical'] += 2
        best = max(scores, key=scores.get)
        return best if scores[best] > 0 else
'worldology'

    def _confidence(self, text: str, domain:
str) -> float:
        words  = set(re.findall(r'\b\w+\b',
text.lower()))
        scores = {d: len(words & kw) for d, kw
```

```
in self.DOMAIN_KEYWORDS.items()}
        total  = sum(scores.values())
        return round(scores.get(domain, 0) /
total, 2) if total > 0 else 0.5

    def _try_compute(self, text: str, domain:
str) -> Optional[str]:
        if domain != 'mathematical':
            return None
        t = text.lower().strip()
        patterns = [
            (r'(?:derivative|differentiate)\s+(.+?)
(?:\s+with respect to\s+(\w+))?$', 'diff'),
            (r'integrate\s+(.+?)(?:\s+with
respect to\s+(\w+))?$', 'integrate'),
            (r'solve\s+(.+?)(?:\s*=\s*0)?$',
'solve'),
            (r'simplify\s+(.+)$', 'simplify'),
            (r'expand\s+(.+)$', 'expand'),
            (r'factor\s+(.+)$', 'factor'),
        ]
        for pat, op in patterns:
            m = re.search(pat, t)
```

```
        if m:
            try:
                expr_str =
m.group(1).strip().replace('^', '**')
                var_str  = (m.group(2).strip()
                            if len(m.groups()) > 1
and m.group(2) else 'x')
                x    = sp.Symbol(var_str)
                expr = sp.sympify(expr_str)
                if op == 'diff':      return
str(sp.diff(expr, x))
                if op == 'integrate': return
str(sp.integrate(expr, x))
                if op == 'solve':     return
str(sp.solve(expr, x))
                if op == 'simplify':  return
str(sp.simplify(expr))
                if op == 'expand':    return
str(sp.expand(expr))
                if op == 'factor':    return
str(sp.factor(expr))
            except Exception:
                pass
```

```
        return None

    def format(self, a: dict, p:
PersonalityEngine) -> str:
        lines = ["Receiver I — Observer/
Analyzer (DMN):"]
        if a['depth'] >= 2:
            lines.append(f"\n  Ontological layer
: {STATE_BRIEF[a['onto_state']]}")
        lines.append(f"  Domain detected   :
{a['domain'].upper()} "
                     f"(confidence:
{a['confidence']:.0%})")
        if a['computed']:
            lines.append(f"\n  Computed result
: {a['computed']}")
        lines.append(f"  Evaluation system :
{a['eval_system']}")
        if a['depth'] == 3:

lines.append(f"\n{STATE_FULL[a['onto_stat
e']]}")
        return p.wrap('\n'.join(lines), 'r1')
```

```
#
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#  RECEIVER II — Decision / Action  (TPN
correlate)
#
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class ReceiverII:

    META_CONTEXT = {
        'mathematical': {'header': "777 Meta-
Decision (Mathematical Domain):",
                         'sin_label': "Avoid in
presenting",
                         'vir_label': "Aim for in the
answer",
                         'rat_label': "Presentation
```

```
strategy"},
        'physical':    {'header': "777 Meta-
Decision (Physical Domain):",
                       'sin_label': "Avoid (false
precision / oversimplification)",
                       'vir_label': "Target (empirical
accuracy + clarity)",
                       'rat_label': "Rigor vs
accessibility balance"},
        'technical':   {'header': "777 Meta-
Decision (Technical Domain):",
                       'sin_label': "Avoid in technical
reasoning",
                       'vir_label': "Standard to aim
for",
                       'rat_label': "Optimal solution
path"},
        'scientific':  {'header': "777 Meta-
Decision (Scientific Domain):",
                       'sin_label': "Avoid (bias /
premature conclusion)",
                       'vir_label': "Prioritize
(evidence, falsifiability)",
```

```
                    'rat_label': "Evidence
synthesis strategy"},
        'worldology':   {'header': "777 Decision
Matrix (Worldology):",
                    'sin_label': "Harmful path to
avoid",
                    'vir_label': "Positive path to
consider",
                    'rat_label': "Optimal
synthesis"},
    }

    SIN_META = {
        'Wrath':    "Avoid aggressive or
overconfident framing.",
        'Greed':    "Avoid overloading with
unnecessary detail.",
        'Sloth':    "Avoid shortcuts that
sacrifice accuracy.",
        'Pride':    "Avoid elegance over clarity.",
        'Lust':     "Avoid chasing complexity for
its own sake.",
        'Envy':     "Avoid unnecessary
```

```
comparisons or competition.",
        'Gluttony': "Avoid committing to too
many approaches at once.",
    }
    VIRTUE_META = {
        'Patience':   "Take the steps needed —
do not rush.",
        'Generosity': "Give enough context for
genuine understanding.",
        'Diligence':  "Check each step carefully
before presenting.",
        'Humility':   "Acknowledge uncertainty
and limits honestly.",
        'Chastity':   "Keep the answer clean,
focused, uncluttered.",
        'Kindness':   "Adapt to the person's
evident level and need.",
        'Temperance': "Balance rigor and
accessibility.",
    }
    RATIONAL_STRATEGY = {
        'Analysis':     "Decompose into clearly
labeled steps.",
```

```
        'Balance':      "Present formal answer
alongside intuition.",
        'Efficiency':   "Lead with the answer;
derivation on request.",
        'Adaptability': "Match depth and
vocabulary to the person.",
        'Prediction':   "Pre-answer likely follow-
up questions.",
        'Optimization': "Shortest correct path;
remove all redundancy.",
        'Resilience':   "If uncertain, state
confidence level explicitly.",
    }

    def decide(self, text: str, domain: str,
depth: int) -> dict:
        idx =
int(hashlib.sha256(text.encode()).hexdiges
t(), 16) % 7
        return {
            'sin': SEVEN_SINS[idx], 'virtue':
SEVEN_VIRTUES[idx],
            'rational': SEVEN_RATIONAL[idx],
```

```
            'context':
self.META_CONTEXT.get(domain,
self.META_CONTEXT['worldology']),
            'sin_advice':
self.SIN_META[SEVEN_SINS[idx]],
            'virtue_advice':
self.VIRTUE_META[SEVEN_VIRTUES[idx]],
            'rational_advice':
self.RATIONAL_STRATEGY[SEVEN_RATION
AL[idx]],
            'domain': domain, 'depth': depth,
        }

    def format(self, d: dict, p:
PersonalityEngine) -> str:
        ctx, depth, domain = d['context'],
d['depth'], d['domain']
        lines = [ctx['header']]
        if depth == 1:
            lines.append(f"  Strategy :
{d['rational']} — {d['rational_advice']}")
        else:
            lines += [
```

```
            f"  {ctx['sin_label']:<42}: {d['sin']}",
            f"    → {d['sin_advice']}",
            f"  {ctx['vir_label']:<42}: {d['virtue']}",
            f"    → {d['virtue_advice']}",
            f"  {ctx['rat_label']:<42}:
{d['rational']}",
            f"    → {d['rational_advice']}",
        ]
        if domain == 'worldology' and depth
>= 2:
            lines += ["",
                  "  60/40 baseline:",
                  "    60% → accuracy /
service to the whole",
                  "    40% → clarity / self-
preservation",
                  "",
                  "  50/50 deadlock → vote or
first instinct",
                  "  Fastest path   → first
instinct immediately"]
        return p.wrap('\n'.join(lines), 'r2')
```

```
#
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#  WORLDOLOGY CONTENT GENERATOR
#  Produces substantive ∞–Unity
responses for human/ethical questions.
#
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class WorldologyResponder:

    def respond(self, text: str, r2: dict) -> str:
        t  = text.lower()
        sn = r2['sin'];   sa = r2['sin_advice']
        vt = r2['virtue']; va = r2['virtue_advice']
        rt = r2['rational']; ra =
r2['rational_advice']
```

```
        if self._has(t, ['consciousness',
'awareness', 'qualia', 'hard problem',
                         'subjective experience', 'what
is mind', 'what is aware']):
            return (
                "Worldology Response —
Consciousness:\n\n"
                "  Within ∞–Unity, consciousness
is not generated by the brain.\n"
                "  It is the pre-existing substrate —
the infinite field (S∞)\n"
                "  that Receiver I filters and
Receiver II acts upon.\n\n"
                "  What you experience as 'your'
awareness is a localized\n"
                "  fragment of the unified field,
shaped by biological limits.\n"
                "    DMN  (Receiver I) : observer
perspective, reflective thought\n"
                "    TPN  (Receiver II): survival,
action, emotional processing\n"
                "  Their synchronization produces
```

```
unified subjective experience.\n\n"
                "  The hard problem dissolves:
qualia are not generated by neurons\n"
                "  — they are filtered
manifestations of the pre-existing field.
\n\n"
                "  50/50 note: This is the
framework's answer.\n"
                "  Whether it is the final truth
remains open —\n"
                "  until personally and directly
experienced."
            )

        if self._has(t, ['choose', 'choice',
'decide', 'decision', 'should i',
                         'what should', 'dilemma',
'which option', 'what do i do',
                         'what to do', 'lost and']):
            return (
                "Worldology Response —
Decision (777 System):\n\n"
                f"  Avoid (sin path) : {sn}\n    →
```

```
{sa}\n"
            f"  Aim for (virtue) : {vt}\n    → {va}
\n"
            f"  Strategy         : {rt}\n    → {ra}
\n\n"
            "  60/40 baseline:\n"
            "    60% → what serves the whole
(others, unity, accuracy)\n"
            "    40% → what preserves you
(clarity, self, boundaries)\n\n"
            "  50/50 deadlock : vote if others
present,\n"
            "                   first instinct if alone.
\n"
            "  Fastest path   : trust first
instinct immediately.\n\n"
            "  The framework gives structure.
You hold the full context."
        )

    if self._has(t, ['forgive', 'relationship',
'trust', 'betray',
                     'hurt me', 'feel lost', 'alone',
```

```
'sad', 'angry',
                    'afraid', 'grief', 'love', 'miss',
'regret',
                    'confused', 'overwhelmed',
'broken']):
        return (
            "Worldology Response —
Emotional/Relational:\n\n"
            "  Within ∞–Unity, difficulty and
pain are not flaws.\n"
            "  They are S2 (duality) doing its
necessary work:\n"
            "  contrast and tension exist so
the infinite can know\n"
            "  itself through constrained, finite
perspective.\n\n"
            "  Your experience — however
difficult — is the infinite\n"
            "  knowing itself from exactly your
vantage point.\n\n"
            f"  777 guidance:\n"
            f"    Avoid : {sn} — {sa}\n"
            f"    Aim   : {vt} — {va}\n\n"
```

```
            "  Practical path (60/40):\n"
            "    60% → what serves the
situation and others\n"
            "    40% → what keeps you whole
and functional\n\n"
            "  The framework offers structure,
not prescriptions.\n"
            "  You hold the lived context — it
does not."
        )

    if self._has(t, ['meaning', 'purpose',
'why am i', 'point of life',
                     'what is life', 'why exist',
'worth living',
                     'what matters', 'significance',
'why are we here']):
        return (
            "Worldology Response —
Meaning/Purpose:\n\n"
            "  The ∞–Unity framework
proposes:\n"
            "  Meaning cannot arise within an
```

```
undifferentiated infinite.\n"
                "  It requires limitation — a finite
perspective within infinity.\n\n"
                "  You are a localized fragment of
the infinite, shaped into\n"
                "  a specific identity so that
infinity can experience itself\n"
                "  from exactly your vantage point
— and only yours.\n\n"
                "  Purpose is not assigned from
outside.\n"
                "  It emerges from the interaction
of your constraints\n"
                "  with the infinite potential you
continuously draw from.\n\n"
                "  Your limitation is not a prison.\n"
                "  It is the instrument through
which the infinite knows itself.\n\n"
                "  50/50 note: This is one
coherent answer. Others exist."
            )

        if self._has(t, ['ethical', 'moral', 'right or
```

```
wrong', 'is it wrong',
                    'is it right', 'harm', 'fair', 'just',
'justice',
                    'good or bad', 'guilty', 'should
i feel']):
        return (
            "Worldology Response — Ethics:
\n\n"
            "  Ethical axioms in ∞–Unity (E1–
E5):\n"
            "    E1: Common origin — all
beings share the same infinite source.\n"
            "    E2: Harm equivalence —
harm(any being) = harm(Unity itself).\n"
            "    E3: Embodied duty — while
present, protect what shares origin.\n"
            "    E5: 777 Protocol — all actions
follow least-harm logic.\n\n"
            "  Applied to your question:\n"
            f"    Avoid : {sn} — {sa}\n"
            f"    Aim   : {vt} — {va}\n"
            f"    Path  : {rt} — {ra}\n\n"
            "  E2 in practice: before any
```

```
action, ask —\n"
                "  does this reduce or increase the
harm-load on Unity?"
            )

        if self._has(t, ['soul', 'death', 'afterlife',
'reincarnation',
                         'what happens when we die',
'after death',
                         'heaven', 'hell', 'spirit',
'immortal']):
            return (
                "Worldology Response — Soul/
Death:\n\n"
                "  Within ∞–Unity, the soul is a
localized piece of the infinite,\n"
                "  shaped into finite identity for
experiential purposes (Ch. 4).\n\n"
                "  Post-mortem (Vol. I, Ch. 6):\n"
                "    Consciousness does not
dissolve — it returns to the substrate.\n"
                "    The form of return is shaped
by the receiver's orientation\n"
```

```
                "    at the moment of
decomposition.\n\n"
                "  This is the framework's
structural answer to continuity.\n"
                "  50/50 note: This remains 50%
real until directly experienced.\n"
                "  The theory does not claim
certainty about post-mortem states."
            )

        if self._has(t, ['free will', 'determinism',
'freedom', 'fate',
                         'destiny', 'predestined', 'do i
have choice', 'am i free']):
            return (
                "Worldology Response — Free Will
/ Determinism:\n\n"
                "  ∞–Unity resolves this via level-
distinction:\n\n"
                "    S2 (local level)  : physical
causation is deterministic.\n"
                "                        Your neurons
follow natural law.\n"
```

```
            "    S∞ (global level) : the infinite
field sustains genuine\n"
            "                        multiplicity and
open trajectories.\n\n"
            "  These do not conflict — they
operate at different layers.\n"
            "  Local determination coexists
with global freedom.\n\n"
            "  In practice (777):\n"
            f"    Act as if you are free — at S∞,
you are.\n"
            f"    Avoid : {sn} — {sa}\n"
            f"    Aim   : {vt} — {va}"
        )

    # Default
    return (
        "Worldology Response — ∞–Unity
Framework:\n\n"
        f"  Ontological layer : S∞\n\n"
        f"  777 guidance:\n"
        f"    Avoid : {sn} — {sa}\n"
        f"    Aim   : {vt} — {va}\n"
```

```
            f"    Path  : {rt} — {ra}\n\n"
            "  60/40: 60% toward the whole, 40%
toward self.\n"
            "  50/50 note: Structure is the gift.
Truth is yours to verify."
        )

    def _has(self, text: str, patterns: list) ->
bool:
        return any(p in text for p in patterns)


#
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#  EXPRESSION PARSER
#  Supports: + - * / ^ and ∞ notation.
Normalizes ÷ × symbols.
#
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```

```
def parse_expression(expr: str):
    e = expr.strip().replace('÷', '/').replace('×',
'*').replace(' ', '')
    m = re.match(
        r'^(-?\d+\.?\d*|inf)([\+\-\*/\^])(-?\d+\.?
\d*|inf)$',
        e, re.IGNORECASE
    )
    if not m:
        return None
    def val(s):
        return inf if s.lower() == 'inf' else
sp.sympify(s)
    return val(m.group(1)), m.group(2),
val(m.group(3))


def fmt(v) -> str:
    return str(v).replace('oo',
'∞').replace('nan', 'NaN')
```

```
#
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#  DUAL-RECEIVER SYSTEM — complete pipeline
#
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class DualReceiverSystem:

    def __init__(self):
        self.core  = OntologicalCore()
        self.r1    = ReceiverI(self.core)
        self.r2    = ReceiverII()
        self.eth   = EthicalCore()
        self.mem   = AdaptiveMemory()
        self.pers  = PersonalityEngine()
        self.wres  = WorldologyResponder()
```

```
    def process(self, text: str) -> str:
        text = text.strip()
        if not text:
            return "  (empty input — awaiting
your question)"

        # Step 0: math expression shortcut
        parsed = parse_expression(text)
        if parsed:
            return self._expr(*parsed)

        # Step 1: ethical gate
        blocked = self.eth.check(text)
        if blocked:
            self.mem.record(text, 'blocked', 1)
            return f"{self.pers.wrap(blocked,
'gate')}\n\n{self.eth.unity_footer()}"

        # Step 2: theory rejection
        if self.eth.check_rejection(text):
            self.mem.rejection_noted = True
            self.mem.record(text, 'rejection', 1)
```

```
            return
self.eth.rejection_response(self.pers)

        # Step 3: personality mode change
        new_mode =
self.pers.detect_and_update(text)
        prefix   = ""
        if new_mode:
            self.mem.record(text, 'personality',
1)
            prefix =
self.pers.transition_message(new_mode)
+ "\n\n"
            # Strip trigger phrase from text
            remaining = text.lower()
            for phrase in
PERSONALITY_TRIGGERS.get(new_mode,
[]):
                remaining =
remaining.replace(phrase, '').strip()
            if len(remaining) < 3:
                return prefix.strip()
            text = remaining
```

```
        # Step 4: depth
        depth = (self.mem.preferred_depth if
self.mem.depth_locked
                 else self._depth(text))

        # Step 5: Receiver I
        r1d  = self.r1.analyze(text, depth)
        dom  = r1d['domain']
        r1_s = self.r1.format(r1d, self.pers)

        # Step 6: Receiver II
        r2d  = self.r2.decide(text, dom, depth)
        r2_s = self.r2.format(r2d, self.pers)

        # Step 7: worldology content
        w_s = None
        if dom == 'worldology':
            w_s =
self.pers.wrap(self.wres.respond(text, r2d),
'w')

        # Step 8: synchronization
```

```
        sync = self._sync(r1d, r2d)

        # Step 9: memory
        self.mem.record(text, dom, depth)

        # Compose
        parts = [r1_s, "", r2_s]
        if w_s:
            parts += ["", w_s]
        if sync and depth >= 2:
            parts += ["", self.pers.wrap(sync,
'sync')]
        parts += ["", self.eth.unity_footer()]

        return prefix + '\n'.join(parts)

    def _expr(self, a, op: str, b) -> str:
        result = self.core.infinity_op(a, b, op)
        is_nan = result is nan or str(result) in
('nan', 'NaN')
        depth  = self.mem.preferred_depth if
self.mem.depth_locked else 2
        key    = f"{a}{op}{b}"
```

```
        if is_nan:
            r1_s = (
                "Receiver I — Observer/Analyzer
(DMN):\n"
                f"  Expression       : {fmt(a)} {op}
{fmt(b)} = NaN\n"
                f"  Ontological layer :
{STATE_BRIEF[OntologicalState.TWO]}\n"
                "  Note             : The one true
undefined in ∞−Unity.\n"
                "                     0 × ∞ — nothingness
meeting infinity.\n"
                "                     Even the dynamic
infinite cannot resolve this."
            )
            r2_s = (
                "777 — NaN / Indeterminate:\n"
                "  This is the only true NaN in ∞−
Unity: 0 × ∞.\n"
                "  Protocol:\n"
                "    With others  → vote, reach
agreement\n"
```

```
            "    Alone        → choose what
feels most coherent\n"
            "    Fastest path → first instinct
immediately"
        )
    else:
        state = self.core.locate(result)
        r1_s = (
            f"Receiver I — Observer/Analyzer
(DMN):\n"
            f"  Expression       : {fmt(a)} {op}
{fmt(b)} = {fmt(result)}\n"
            f"  Ontological layer :
{STATE_BRIEF[state]}\n"
            "  ∞–Unity semantics : Infinity is
dynamic — it cannot be diminished.\n"
            f"  ({fmt(a)} {op} {fmt(b)} = ∞
because ∞ op ∞ = ∞ in this framework)"
        )
        r2d  = self.r2.decide(key,
'mathematical', depth)
        r2_s = self.r2.format(r2d, self.pers)
```

```
        self.mem.record(key, 'mathematical',
1)
        return (f"{self.pers.wrap(r1_s, 'r1')}
\n\n"
                f"{self.pers.wrap(r2_s, 'r2')}\n\n"
                f"{self.eth.unity_footer()}")

    def _depth(self, text: str) -> int:
        w = set(text.lower().split())
        if
{'prove','derive','formalize','rigorous','theorem'
,

'detailed','complete','explain','elaborate'} & w
or len(text.split()) > 25:
            return 3
        if {'tldr','briefly','summary','quick'} & w:
            return 1
        return 2

    def _sync(self, r1: dict, r2: dict) -> str:
        dom, strat, onto = r1['domain'],
```

```
r2['rational'], r1['onto_state']
        if dom != 'worldology':
            return (
                "Synchronization (Receiver I ↔
Receiver II):\n"
                f"  Domain    : {dom}\n"
                f"  Strategy  : {strat} — 777
optimizes delivery, not content\n"
                "  Principle : WHAT = formal truth
criteria | HOW = 777 logic"
            )
        return (
            "Synchronization (Receiver I ↔
Receiver II):\n"
            f"  Layer     : {STATE_BRIEF[onto]}\n"
            "  Mode      : 777 is the object-level
decision\n"
            "  Status    : Both receivers active —
qualia-level analysis engaged\n"
            "  50/50     : Framework gives
structure; lived context is yours."
        )
```

```
#
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#  MAIN LOOP
#
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HELP_TEXT = """
  WORLDOLOGY ∞–Unity Theory —
Commands

_______________________________________________________________
_______________________________________________________________
_____________________

  Any question    → processed through dual-
receiver pipeline
  inf±*/^n        → ∞–Unity arithmetic  (inf-inf
 5*inf  2^10)
```

Commands:
status → current mode, depth, ethical state
profile → session statistics
history → last 5 inputs
continuum → full ontological continuum (S0 → S1 → S2 → S∞)
depth 1/2/3 → lock output depth
depth auto → adaptive depth (default)
help → this message
exit → end session

Personality (phrase-triggered, not keyword):
"be funny" → comedy mode
"be socratic" → Socratic mode
"be brief" → minimal mode
"be poetic" → poetic mode
"reset personality" → standard mode

Examples:
"I don't know what to choose"

```
    "what is consciousness"
    "derivative of x**3 + sin(x)"
    "inf - inf"   "0*inf"   "inf/inf"
    "I reject this theory"

_____________________________________________________
_____________________________________________________
_____________________
"""

SEP  = "═" * 66
SEP2 = "─" * 66


def main():
    sys_ = DualReceiverSystem()

    print(SEP)
    print(f"  WORLDOLOGY — ∞–Unity
Theory  ({VERSION})")
    print(f"  Dual-Receiver Simulation  |
Architecture {VERSION}")
    print(f"  Philosophy: Taha Givarian  |
```

```
Code: AI  |  Feb 2026")
    print(SEP)
    print()
    print("  Receiver I  (DMN) → Observer,
domain analysis, formal systems")
    print("  Receiver II (TPN) → 777 decision,
always active, all domains")
    print()
    print("  ∞–Unity arithmetic: ∞ op ∞ = ∞  |
Only NaN: 0 × ∞")
    print("  Type 'help' for all commands.")
    print(SEP)

    while True:
        try:
            raw = input("\n> ").strip()
        except (EOFError, KeyboardInterrupt):
            print("\n  Cycle complete. Energy
returns to Unity.")
            break

        if not raw:
            continue
```

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        cmd = raw.lower()

        if cmd == 'exit':
            print("  Cycle complete. Energy
returns to Unity.")
            break

        if cmd == 'help':
            print(HELP_TEXT)
            continue

        if cmd == 'status':
            m, p, e = sys_.mem, sys_.pers,
sys_.eth
            print(f"\n  Ethical state    :
{e.state.value.upper()}")
            print(f"  Personality mode :
{PERSONALITY_LABELS[p.mode]}")
            print(f"  Depth mode       :
{m.depth_label()} "
                  f"({'locked' if m.depth_locked
else 'adaptive'})")
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            print(f"  Turn count      :
{m.turn_count}")
            print(f"\n{e.unity_footer()}")
            continue

        if cmd == 'profile':
            print(f"\n{sys_.mem.profile()}")
            continue

        if cmd == 'history':
            print(f"\n{sys_.mem.last_inputs(5)}")
            continue

        if cmd == 'continuum':
            print()
            for state in OntologicalState:
                print(STATE_FULL[state])
                print()
            continue

        if cmd.startswith('depth '):
            arg = cmd.split(None, 1)[1]
            if arg == 'auto':
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            sys_.mem.depth_locked    = False
            sys_.mem.preferred_depth = 2
            print("  Depth → adaptive (auto).")
        elif arg in ('1', '2', '3'):
            sys_.mem.preferred_depth =
int(arg)
            sys_.mem.depth_locked    = True
            print(f"  Depth locked → {arg}
({sys_.mem.depth_label()}).")
        else:
            print("  Usage: depth 1 / depth 2 /
depth 3 / depth auto")
        continue

    print()
    print(sys_.process(raw))
    print(SEP2)


if __name__ == "__main__":
   main()
```